# MILITARY SPECIFICATION

## PROJECTILE, 20MM, HEI, M56A4; METAL PARTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the projectile metal parts for Projectile, 20MM, High Explosive Incendiary, M56A4, manufactured by the Blank, Cup and Draw Process.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| TT-C-490 | - | Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings |

MILITARY

| | | |
|---|---|---|
| MIL-A-48078 | - | Ammunition, Standard Quality Assurance Provisions, General Specifications for |

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-651 | - | Visual Inspection Standards for 20MM Ammunition and Components |
| MIL-STD-1168 | - | Lot Numbering of Ammunition |
| MIL-STD-1169 | - | Packaging, Packing and Marking for Shipment of Inert Ammunition Components |

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.8)

U.S. Army Armament Research & Development Center (ARDC)

Product and Packaging Drawings

| | | |
|---|---|---|
| 7258816 | - | Blank, Rotating Band |
| 7259136 | - | Shipping Container for Inert Projectile, 20MM Assemblies and Metal Parts Assemblies (10,000) |
| 7259231 | - | Projectile, 20MM, HEI, M56A4, Metal Parts |
| LI 7259231 | - | Index of Inspection Equipment Lists for: Projectile, 20MM, HEI, M56A4, Metal Parts |
| 7259232 | - | Body and Rotating Band Assembly, Projectile, 20MM, HEI, M56A4 |
| 7259233 | - | Body, Projectile, 20MM, HEI, M56A4 |
| 7258861 | - | Carton, for Interplant Shipment of Projectiles, 20MM, Assemblies and Metal Parts Assemblies (200) |
| 7259087 | - | Cartridge, 20MM, for HEI Projectile Metal Parts Test |



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Inspection Equipment Drawings

| | | |
|---|---|---|
| 7259353 | - | Go, Flat Plug Gage, Special |
| 7259419 | - | Band Tightness Tester |
| 7259433 | - | Gage Indicator |
| 7259434 | - | Scale |
| 7259467 | - | Comparator, Optical, Hand Type |
| 7259489 | - | Adapter, Thread Plug |
| 7259492 | - | Acceptance Check Gage |
| 7259497 | - | Go, Thread Plug Gage |
| 7259536 | - | Not Go, Flat Plug Gage, Special |
| 7259545 | - | Operating Instructions (Band Tester) |
| 7259678 | - | Gage, Indicator |
| 7259718 | - | Optical Gaging Chart |
| 7259740 | - | Gage, Snap, Special |
| 8655592 | - | Go, Ring Gage, Special |
| 8657201 | - | Optical Gaging Chart |
| 9332527 | - | Ultrasonic Standards Probe Set-up for M56A4 Projectile Body |
| 11075081 | - | Gage, Indicator |
| 11075092 | - | Gage, Ring, Plain, Go, Special |
| 11075094 | - | Gage, Snap, Go, Special |
| 11075203 | - | Eddy Current Flaw Detector |
| 11075239 | - | Gage, Comparator |
| 11075247 | - | Requirements for Optical Gaging Charts |
| 11075248 | - | Block, Setting |
| 11075270 | - | Optical Gaging Chart |

PUBLICATIONS

| | | |
|---|---|---|
| AMCR 715-505 | - | Ammunition Ballistics Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges. |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Tesing and Materials (ASTM)
ASTM B117 - Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

2.1.4 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

a Material. Materials shall be in accordance with the applicable drawings and specifications.

b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (6.5).

3.2 Components. All parts and assemblies shall comply with all requirements specified on Drawing (Dwg.) 7259231, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article sample. Unless otherwise directed by the contracting officer, a first article sample shall be required.

3.4 Laminar metallic structure. Each projectile body shall be free of laminar metallic structures as detectable by ultrasonic testing methods approved by the contracting officer.

3.5 Banding. The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the rotating band seat after knurling. The rotating band blank shall fill the band seat.



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3.6 Soundness. Metal discontinuity in the projectile body, as determined by eddy current inspection, shall be less than that represented by Production Check Standard (PCS) No. 1.

3.7 Protective coating.

3.7.1 Phosphate coating. The pretreatment process used shall be in accordance with applicable requirements of Dwg. 7259231. The phosphate coating shall be in accordance with the requirements of TT-C-490 for appearance and minimum weight, as applicable to the type of phosphate coating used. Test specimens (panels) shall be in accordance with the requirements of TT-C-490.

3.7.2 Paint coating. The interior and exterior coatings of the projectile metal parts shall be in accordance with applicable requirements of Dwg. 7259231. The exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490.

3.7.3 Corrosion resistance. The exterior painted surfaces of the projectile metal parts shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.8 Metal parts security. The projectile metal parts, inert charged, and assembled into cartridges in accordance with Dwg. 7259087, shall not separate or break up in the test for metal parts security.

3.9 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Processing defects. The projectile metal parts shall be free of defects, such as dents, deep scratches, cuts, excessive surface roughness, metal flaws, burrs, sharp edges, and other forms of substandard workmanship.

3.9.2 Cleanliness. The projectile metal parts shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.



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4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (4.3)
b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

a. Inspection and nonfiring tests:

| Part Description | Drawing | Quantity | Method Para |
|---|---|---|---|
| Standard test panels | | 3 | 4.5.7 |
| Rotating Bands | 7258816 | 32 | 4.4.2.1 |
| Projectile bodies | 7259233 | 125 | 4.4.2.2 |
| Projectile body assemblies (uncoated) | 7259232 | 125 | 4.4.2.3 |
| Projectile metal parts (coated prior to paint) | 7259231 | 125 | 4.4.2.4 |
| Projectile metal parts (painted) | 7259231 | 170 | 4.4.2.5 |

b. Firing test:

| Parts description | Drawings | Quantity | Method Para |
|---|---|---|---|
| Projectile, metal parts | 7259231 | 4,000 | 4.5.10 |

c. The projectile metal parts, Dwg. 7259231, shall be identified in accordance with MIL-STD-1168.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with the applicable drawings and requirements of Table I, as specified herein, will result in sample disapproval.



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TABLE I. FIRST ARTICLE INSPECTION

**CLASSIFICATION OF DEFECTS & TESTS** MIL-P-60295C

| PARAGRAPH | TITLE | | | SHEET OF | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Projectile, 20MM, HEI, M56A4, Metal Parts | | | 1 2 | SEE BELOW |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Rotating Bands - Dwg. 7558816 | | | | |
| | Examination for defects | 32 | | 3.2 | 4.4.2.1 |
| | Body, Projectile - Dwg. 7259233 | | | | |
| | Examination for defects | 125 | | 3.2 | 4.4.2.2 |
| | Laminar metallic structure | 125 | 0-1* | 3.4 | 4.5.1 |
| | Body and Rotating Band Assembly (Uncoated) - Dwg. 7259232 | | | | |
| | Examination for defects | 125 | | 3.2 | 4.4.2.3 |
| | Soundness verification | 125 | 0-1* | 3.6 | 4.5.6 |
| | Projectile, Metal Parts (Coated Prior to Paint) - Dwg. 7259231 | | | | |
| | Examination for defects | 125 | | 3.2 | 4.4.2.4 |
| | Standard test panels (specimens) | 3 | 0-1* | 3.7.1 | 4.5.7 |

NOTES:

*Accept - Reject Criteria.



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TABLE I. FIRST ARTICLE INSPECTION

**CLASSIFICATION OF DEFECTS & TESTS** MIL-P-60295C

| PARAGRAPH | TITLE | | SHEET OF | | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Projectile, 20MM, HEI, M56A4, Metal Parts | | 2 2 | | SEE BELOW |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Projectile, Metal Parts (Painted) | | | | |
| | Dwg. 7259231 | | | | |
| | Examination for defects | 125 | | 3.2 | 4.4.2.5 |
| | Band tightness | 125 | 0-1* | 3.5 | 7259419/4.5.3 |
| | Paint adhesion | 13 | 0-1* | 3.7.2 | 4.5.8 |
| | Corrosion resistance | 32 | 0-1* | 3.7.3 | 4.5.9 |
| | Firing Test | | | | |
| | Metal parts security Test | 4000 | 0-1* | 3.8 | 4.5.10 |

NOTES:

*Accept - Reject Criteria.

DRSMC 9A (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. To determine product acceptability, major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.50 percent and the AQL for minor defects shall be 2.50 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, when three or less defects appear in a classification, sampling plans applicable to a class shall not be used for major defects. (See 6.7)

c. MIL-STD-651 shall apply in defining and evaluating visual defects.



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60295C

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 7258816 |
|---|---|---|---|---|---|
| 4.4.2.1 | Rotating Band Blank | | | | NEXT HIGHER ASSEMBLY 7259232 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Foreign matter | | .25% | 3.9.2 | Visual |
| 102. | Defective metal, including lamination | | .25% | 3.2 | Visual |
| 103. | Incomplete or deformed | | .25% | 3.2 | Visual 1/ |
| MINOR | | | | | |
| 201. | Width, max. | | .40% | 3.2 | 11075094 |
| 202. | Evidence of poor workmanship | | .65% | 3.9 | Visual |

NOTES: 1/ A rotating band blank which is deformed or is judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions, shall be classed defective.



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# CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60295C

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.2 | Body, Projectile | 1 OF 1 | 7259233 |
| | | | NEXT HIGHER ASSEMBLY: 7259232 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | | | | | |
| 1. | Laminar metallic structure | | 100% | 3.4 | 4.5.1 |
| MAJOR | | | | | |
| 101. | Length, mouth to rear edge of rotating band seat | | .25% | 3.2 | 7259740 |
| 102. | Knurling missing or incomplete | | .25% | 3.2 | Visual |
| 103. | Depth, knurling, min. | | .25% | 3.2 | Visual/7259678 1/ |
| 104. | Foreign matter in rotating band seat | | .25% | 3.5 | Visual |
| MINOR | | | | | |
| 201. | Width, rotating band seat | | .40% | 3.2 | 7259353/7259536 |
| 202. | Diameter, rotating band seat knurling | | .40% | 3.2 | Gage 2/ |
| 203. | Width, knurling, min. | | .40% | 3.2 | Visual/7259467 1/ |
| 204. | Evidence of poor workmanship | | .65% | 3.9 | Visual |

NOTES:

1/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

2/ Gage, Snap - (Anvils, .170 min. width)
(Go .680) .6796 - .0004
(Not Go .662) .6620 + .0004



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60295C

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.3 | Body and Rotating Band Assembly, Uncoated | 1 OF 3 | 7259232 |
| | | | NEXT HIGHER ASSEMBLY 7259231 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | | | | | |
| 1. | Soundness-Ferrous Metal Discontinuity Test | | 100% | 3.6 | 4.5.4/11075203 |
| MAJOR | | | | | |
| 101. | Diameter, bourrelet min. | | .25% | 3.2 | Gage 1/ |
| 102. | Diameter, body, rear of rotating band, min. | | .25% | 3.2 | Gage 2/ |
| 103. | Run out, body, rear of rotating band, with datum surface | | .25% | 3.2 | 7259433/7259492/ 7259489 |
| 104. | Diameter, rotating band | | .25% | 3.2 | Gage 3/ |
| 105. | Length, mouth to rear edge of rotating band, max. | | .25% | 3.2 | 11075239/11075248 |
| 106. | Run out, rotating band with datum surface | | .25% | 3.2 | 7259433/7259492/ 7259489 |
| 107. | Length, to internal basic diameter | | .25% | 3.2 | Gage |

NOTES:

1/ Snap Gage - (Not Go .782) .7820 + .0001
2/ Snap Gage - (Not Go .765) .7650 + .0003
3/ Ring Gage - (Go .828) .82780 - .00012
Snap Gage - (Not Go .824) .8240 + .0002



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60295C

| PARAGRAPH | TITLE | | SHEET OF | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.3 | Body and Rotating Band Assembly, Uncoated | | 2 3 | | 7259232 |
| | | | | | NEXT HIGHER ASSEMBLY 7259231 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| 108. | Profile, crimp groove | | .25% | 3.2 | Comparator/20x/8657201 |
| 109. | Radius, rear edge of crimp groove, max. | | .25% | 3.2 | Comparator/50x |
| 110. | Location, crimp groove | | .25% | 3.2 | Comparator/20x/8657201 |
| 111. | Wall thickness variation, max. | | .25% | 3.2 | Gage |
| 112. | Thickness, base, min. | | .25% | 3.2 | 11075081 4/ |
| 113. | Thickness, wall, min. | | .25% | 3.2 | Gage |
| 114. | Total length, max. | | .25% | 3.2 | Gage |
| 115. | Void, junction rotating band and body | | .25% | 3.5 | Visual |
| 116. | Threads not complete | | .25% | 3.2 | Visual |
| 117. | Base face machined | | .25% | 3.2 | Visual |
| 118. | Radius, base of projectile | | .25% | 3.2 | Visual/Comparator/50x 5/ |
| MINOR | | | | | |
| 201. | Depth, thread counterbore (if present), max. | | .40% | 3.2 | Visual/Gage 6/ |
| 202. | Diameter, thread counterbore or chamfer (if present) | | .40% | 3.2 | Visual/Gage 6/ |

NOTES:

4/ Base thickness shall be measured at the longitudinal axis from opposite surface flats. A body having base thickness less than the specified minimum shall be classed defective.

5/ Measurement of mouth-ogive break and the base radius shall apply where visual inspection indicates that the maximum dimension is borderline.



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**CLASSIFICATION OF DEFECTS & TESTS** MIL-P-60295C

| PARAGRAPH | TITLE | | SHEET OF | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.3 | Body and Rotating Band Assembly, Uncoated | | 3 3 | | 7259232 |
| | | | | | NEXT HIGHER ASSEMBLY 7259231 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| 203. | Outside diameter, mouth end | | .40% | 3.2 | 11075270/Comparator/ 10x |
| 204. | Total length, min. | | .40% | 3.2 | Gage |
| 205. | Length, mouth to rear edge of rotating band, min. | | .40% | 3.2 | 11075239/11075248 |
| 206. | Finish improper | | .40% | 3.2 | Visual |
| 207. | Break, mouth, at ogive, max. | | .40% | 3.2 | Visual/11075270/ Comparator/10x 5/ |
| 208. | Depth and width of undercut front and rear of rotating band | | .40% | 3.2 | 7259718/Comparator/ 31.25x |
| 209. | Mutilated | | .40% | 3.2 | Visual |
| 210. | Corrosion | | .40% | 3.2 | Visual |
| 211. | Width, rotating band | | .40% | 3.2 | Comparator/20x/8657201 |
| 212 | Evidence of poor workmanship | | .65% | 3.9 | Visual |

NOTES: 6/ Measurement of the diameter and depth, max., of the thread counterbore or chamfer shall apply where visual inspection indicates that the diameter or depth max., dimension is borderline.

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60295C

| PARAGRAPH | TITLE | SHEET OF | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.4 | Projectile, Metal Parts (Coated, Prior to Painting) | 1 1 | 7259231 |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Pitch diameter, thread | | .25% | 3.2 | 7259497 1/ |
| 102. | Run out, thread with datum surface | | .25% | 3.2 | 7259433/7259492/7259489 |
| MINOR | | | | | |
| 201. | Minor diameter, thread | | .40% | 3.2 | Gage 2/ |
| 202. | Length, complete thread, min. | | .40% | 3.2 | Gage 1/ |
| 203. | Evidence of poor workmanship | | .65% | 3.9 | Visual |

NOTES:

1/ Thread Plug Gage - Go .5422 & thread length, min.
Thread Plug Gage, expanding type (cone and vee) w/stop*, Not Go .5469
Thread Set Check - Thread Ring Gage (.5469 - .0001)

2/ Plain Plug Gage, expanding type w/stop*
.5287 + .0057 (max. anvil width .080)
Thread Set Check, Plain Ring Gage (.53155 ± .00005)

* Stop shall be suitable to check thread throughout min. thread depth specified on component drawing.



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60295C

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 7259231 |
|---|---|---|---|---|---|
| 4.4.2.5 | Projectile, Metal Parts (Painted) | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Diameter, bourrelet, max. | | .25% | 3.2 | 8655592 |
| 102. | Diameter, body, rear of rotating band, max. | | .25% | 3.2 | 11075092 |
| 103. | Weight | | .25% | 3.2 | Balance/7259434 |
| 104. | Foreign matter in cavity or threads | | .25% | 3.2 | Visual 1/ |
| MINOR | | | | | |
| 201. | Interior or exterior paint inadequate | | .40% | 3.2 | Visual |
| 202. | Exterior paint not dry | | .40% | 3.2 | Manual |
| 203. | Paint on rotating band | | .40% | 3.2 | Visual |
| 204. | Paint on threads | | .40% | 3.2 | Visual |
| 205. | Marking incorrect, incomplete, illegible or missing | | .40% | 3.2 | Visual |
| 206. | Rotating band damaged | | .40% | 3.2 | Visual |
| 207. | Foreign matter on exterior | | .40% | 3.2 | Visual |
| 208. | Evidence of poor workmanship | | .65% | 3.9 | Visual |

NOTES:

1/ Particular attention shall be given to examination of the cavity for steel chips and shavings.



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4.4.3 Tests. The tests listed in Table II shall be conducted on each lot in compliance with the methods of inspection prescribed in 4.5. The samples shall be selected in such a manner as to be representative of the entire lot. The sample size and acceptance criteria for each test shall be as specified.

TABLE II

| Test | | Sample Size | Item | Req. Para |
|---|---|---|---|---|
| Laminar metallic structure verification | 1/ | 315 | Body, after tapering operation (See 4.5.2) | 3.4 |
| Band tightness | 2/ | 315 | Projectile, metal parts (See 4.5.3) | 3.5 |
| Soundness verification | 3/ | 315 | Projectile, metal parts, unpainted (See 4.5.6) | 3.6 |
| Automatic inspection effectiveness | 4/ | | PCS No. 1 (See 4.5.5) | |
| Phosphate coating (Appearance & weight) | 5/ | 3 | Projectile, metal parts (before painting/or standard panels) (See 4.5.7) | 3.7.1 |
| Paint adhesion | 6/ | 13 | Projectile, metal parts, TT-C-490 (See 4.5.8) | 3.7.2 |
| Corrosion resistance | 7/ | 32 | Projectile, metal parts (See 4.5.9) | 3.7.3 |
| Metal parts security | 8/ | 250 | Projectile, metal parts (See 4.5.10) | 3.8 |

1/ Failure of one or more units of the sample to comply with the applicable requirements for laminar metallic structure shall be cause for rejection of the lot.

2/ If two or more units of the sample have an average radial void between the seated band and the band seat, greater than 0.004 inch, the lot shall be rejected. If one unit of the sample has an average radial void, greater than 0.004 inch, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units have an average radial void greater than 0.004 inch, the lot shall be rejected.

3/ Rejection of one or more units of the sample by the flaw detector in the test for body metal soundness shall be cause for rejection of the lot.



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4/ Failure of the flaw detector to reject PCS No. 1 shall be cause for rejection of all projectile metal parts inspected for "soundness" since the preceding test and shall require requalification of the equipment for automatic inspection. (See 4.5.5)

5/ Failure of the phosphate coating on the sample projectiles or standard panels to comply with the applicable requirements shall cause rejection of all production projectile metal parts phosphated since the preceeding test. (See 3.7.1)

6/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.

7/ If two or more units of the sample fail to comply with the corrosion resistance requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

8/ The lot will be rejected, if any one of the following types of metal parts separation or breakup occurs in the metal parts security test, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s).

a. Body breakup.

b. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

c. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

d. The following acceptance criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint.



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(1) The lot will be accepted if two or less projectile metal parts throw band fragments having length not greater than 0.500 inch.

(2) The lot will be rejected if three or more projectile metal parts throw band fragments having length not greater than 0.500 inch.

(3) The lot will be rejected if any projectile metal parts throws a band fragment having length greater than 0.500 inch (indicative of deep-seated band lamination).

(4) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

4.4.4 Packaging, packing and marking. Inspection for packaging, packing and marking shall comply with applicable requirements of MIL-STD-1169.

4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Index of Inspection Equipment List-LI 7259231 and the Paragraph Reference/Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Laminar metallic structure. The contractor shall develop an ultrasonic inspection technique/procedure to detect effectively laminar metallic structure in the projectile body as detailed on Dwg. No. 9332527. The contractor shall obtain technical agency approval for incorporation into the inspection system.

4.5.1.1 Application of defect standards. Government designed standards will be employed to set-up, calibrate and periodically check the performance of the ultrasonic equipment, (see 6.6.2). Set-up, calibration, and periodic check procedures are to be included as part of the ultrasonic inspection procedures to be submitted to the Government for approval.

4.5.1.2 Equipment verification. The Government inspector will maintain independent of the contractor, a set of standards and may cycle these standards through the equipment at any time. Failure of equipment to reject the standards will be evidence that the equipment is not properly adjusted and the contractor will be so notified.



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4.5.2. Laminar metallic structure verification. The method of test shall be in accordance with 4.5.1. A body having a flaw indication shall be sectioned normal to and at the midpoint of the indication. The sectioned specimen shall be etched, polished and examined at 20 diameters. An indication which is shown to be laminar metallic structure shall cause the body to be classed defective.

4.5.3 Band tightness. The method of test shall comply with all requirements of the operating instructions for the band tightness tester Dwg. 7259545. The inspection sample shall be cycled through the tester to determine compliance with the rotating band tightness requirement. The total measurement of radial void registered on the tester indicator at final ram pressure, shall be divided by four (number of indentors) to determine the average radial void.

4.5.4 Soundness. The method of inspection shall comply with all requirements of Eddy Current Flaw Detector, Dwg. 11075203, except for those inspection requirements which are reserved for Government application, as specified herein.

4.5.4.1 Application of defect standards. Master defect standards, referenced in the Operating Instructions, shall apply to resolve inspection discrepancies occurring between the results of defect standards supplied for contractor "soundness" inspection and those supplied for Government "soundness" verification. (See 6.6)

4.5.5 Automatic eddy current inspection effectiveness. Equipment checkout shall be performed as stated in the operating instructions at the start of each shift and after any adjustment or shutdown. Upon completion of checkout, cycle the PCS No. 1, front slot and rear slot standard through the inspection system during automatic operation. All three standards must be rejected to assure correct adjustment for reject level and scan limits. During remainder of shift the PCS No. 1 shall be cycled through the equipment at intervals of 20 $\pm$ 5 minutes during automatic operation.

4.5.5.1 Equipment verification. The Government inspector will maintain independent of the contractor, a set of standards and may cycle these standards through the equipment at any time. Failure of equipment to reject standards will be evidence that the equipment is not properly adjusted and the contractor will be so notified.



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4.5.6 Soundness verification. After equipment checkout and verification as specified in 4.5.5 and 4.5.5.1 the entire lot sample will be inspected with the equipment operating automatically. Any rejected unit of the sample will be automatically cycled through the equipment with the CAL. button pressed (to lower the sensitivity one step below that used for production inspection). Any unit which is rejected at the lower sensitivity shall be classed as a defective.

4.5.7 Phosphate coating. The method of test shall be as prescribed in TT-C-490 for the type of coating used (see Dwg. 7259231). The prescribed sample projectiles or standard panels shall be processed at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.5.8 Paint adhesion. The method of test shall be as prescribed in TT-C-490. The test shall be made on the exterior painted surface of each projectile metal part of the sample forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal on any unit of the sample, as a result of the test, shall be recorded as a defect.

4.5.9 Corrosion resistance. The method of test shall be as prescribed in American Society for Testing and Materials (ASTM), ASTM B117. The paint shall not be disturbed, however, the rotating band and unpainted surfaces of the sample projectile metal part may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.7.3.

4.5.10 Metal parts security. The method of test will be as prescribed in AMCR 715-505, Volume 8, as applicable. The sample projectile metal parts will be loaded and assembled as specified on Dwg. 7259087, Cartridge, 20MM, for projectile metal parts test. The test cartridges, at ambient temperature, will be fired in bursts of 50 shots in an M39 gun. The gun barrel will be at ambient temperature at the beginning of the test and cooled to ambient temperature after each 100 rounds fired. The cyclic rate of fire will be recorded for informational purposes.

## 5. PACKAGING

### 5.1 Preservation and Packaging.

5.1.1 The projectile metal parts shall be preserved and packaged in accordance with Dwg. 7258861 or Dwg. 7259136, as directed by the procuring activity.



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5.2 Packing.

5.2.1 The projectile metal parts shall be preserved and packaged as specified in 5.1. Each lot shall be packed for shipment in accordance with Dwg. 7258861 or Dwg. 7259136, as directed by the procuring activity.

5.3 Marking.

5.3.1 Marking or labeling of packed cartons shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use with Cartridges, 20MM, HEI, M56A3 and M56A4.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRSMC-QAF-I(D), US Army Armament Research and Development Center (ARDC), Dover, NJ, 07801.

6.4 First article sample. Where a first article sample is required by the contract, the contracting officer will notify the test facility when to proceed with the metal parts security test. Additional assemblies may be required by the test facility. (See 4.3)

6.5 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.6 Defect standard.

6.6.1 Defect standard for eddy current testing. The initial supply of defect standards and instructions for contractor and government application to the eddy current method of inspection will be furnished. Replacement standards should be requisitioned through normal channels by the Government agency having contract administration responsibility.

6.6.2 Defect standards for ultrasonic testing. The contractor will be responsible for manufacture and maintenance of the appropriate defect standards for the ultrasonic testing.



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6.7 Intermediate point inspection. The classification of defects identifies the defect characteristics (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which would be detrimental to the attainment of optimum product quality of the end item.

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, or US Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

6.9 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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